

Tasks



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Section Introduction-Tasks

- What are Tasks ? Their benefits and limitations.
- Task features and how you can use them ?
- LAB
 - Create, Alter, Schedule and Execute Tasks.
 - Create Tasks without an execution schedule or Warehouse.
 - Create serverless Tasks without a Warehouse
 - Use more than 1 SQL statement in a Task.
 - Use Conditions with Tasks.
 - Create a Tree of Tasks to control Task execution order.
 - Locate Task dependency.
 - Understand Finalize Task.
 - Monitoring Tasks.
 - Use Merge and Copy Command with Tasks.
- Summary.

What are Tasks?

Tasks are Snowflake's implementation of job scheduling with which we can run DML statements like `INSERT, UPDATE and DELETE`, DDL statement like `MERGE, TRUNCATE, CREATE` and `DROP` or invoke stored procedures with the `CALL` statement based on a schedule .

Benefits of Tasks

- Tasks eliminate the need to manually run SQL statements improving productivity and reducing depending on external schedules like Airflow, Control-M and Tivoli.
- Tasks simplify scheduling as they can be run at cyclic intervals, We can schedule tasks to run every 1 min, 30 mins, 60 mins etc. Eg: `SCHEDULE = '1 minute'`.
- Tasks can be scheduled using cron . Eg: `SCHEDULE = 'USING CRON 0 7* *1-5 America/Chicago'`
- Tasks can poll a Stream to check data and execute SQL statements if there is data in it.
- We can create multiple Tasks and dependency between them to control the order of Task execution. Creates parent child relationship called DAG, Tree of Tasks or Hierarchy of Tasks.

Limitations of Tasks

- Tasks cannot create dependency with external systems.
- Task can have a maximum of 1 SQL statement associated with it.
- Tasks can run for a maximum of 60 mins after which it will be aborted.
- A DAG which is a tree of Tasks can have a maximum of 1000 Tasks including the root Task.
- A Task can have a maximum of 100 child Tasks and 100 parent Tasks.

Tasks Features

- When a Task is created it is inactive by default.
- We can enable a task by using `ALTER TASK<taskname> RESUME`.
- We can disable or suspend a TASK by using `ALTER TASK<taskname> SUSPEND`.
- Tasks can be created with or without a WAREHOUSE (After Sept 2021)
- Snowflake uses `MEDIUM` warehouse as the default size for serverless Tasks.
- Provide optional parameter `USER_TASK_MANAGED_INITIAL_WAREHOUSE_SIZE` which will be the starting compute size for serverless Tasks to override the default of Medium.
- Task can be created by using the

```
CREATE OR REPLACE TASK <taskname-one>
WAREHOUSE =COMPUTE_WH
SCHEDULE =1 MINUTE
AS <SQL Statement>
```
- Dependencies also known as parent child relationships between Tasks can be created by using the `AFTER` keyword

```
CREATE OR REPLACE TASK <taskname-two>
WAREHOUSE =COMPUTE_WH
AFTER <taskname-one>
AS <SQL Statement>
```
- Use the `ALTER TASK <taskname-two> REMOVE AFTER <taskname-one>` to remove dependencies between Tasks.

Tasks Features

- When a Task is suspended, dependent child Tasks will not execute.
- Tasks can be monitored by using the [INFORMATION_SCHEMA.TASK_HISTORY](#) to see the the execution status of Tasks , data is available for 7 days.
- Task data more than 7 days old can be found in [ACCOUNT_USAGE.TASK_HISTORY](#), there is a 45 min lag before Task data shows up in [ACCOUNT_USAGE.TASK_HISTORY](#) and hence it is not used for monitoring Tasks.
- Execution information is available immediately in [INFORMATION_SCHEMA.TASK_HISTORY](#) and should be used for monitoring Tasks.
- Dependency information between Tasks can be found in [INFORMATION_SCHEMA.TASK_DEPENDENTS](#).
- Tasks can be manually executed with the [EXECUTE TASK <task-name>](#).
- Task can be dropped by using [DROP TASK <task-name>](#).
- List can be generated by using [SHOW TASKS](#).
- We can describe a Task with [DESCRIBE TASK <task-name>](#) to see the properties of a task including if it is in suspended state or not.
- We can view, manage and see execution information on Tasks from Snowsight and visually see dependencies between parent and child Tasks.
- Tasks are automatically suspended when a Task fails 10 times this value can be changed by updating the [SUSPEND_TASK_AFTER_NUM_FAILURES](#) parameter.

Tasks Features

- Tasks do not allow overlapping executions by default, this is the default behavior and can be changed by updating the [ALLOW_OVERLAPPING_EXECUTION](#) parameter.
- When a Task fails you can get snowflake to automatically retry the failed Tasks. This is turned off by default and can be enabled by using [TASK_AUTO_RETRY_ATTEMPTS](#) parameter and must be set at the root Task level.



Summary

- Tasks are implementation of scheduling in Snowflake.
- Snowflake supports 2 types of data scheduling approach. Cyclic scheduling in which we provide a **SCHEDULE** interval like 1 minute, 10 minute(s) or **CRON** where we provide information on when the job should run.
- Tasks are in suspended state when created, they need to be resumed to be able to run.
- We can create a Task without a warehouse, Tasks will use serverless compute to run if they are created without a warehouse.
- **USER_TASK_MANAGED_INITIAL_WAREHOUSE_SIZE** can be provided optionally to help Snowflake start with an appropriate warehouse size if a serverless Task is being created.
- Providing a schedule to a Task is optional, if a schedule is not provided the Task will not run and will need to be manually executed with the **EXECUTE TASK <taskname>**
- Tasks can execute only 1 SQL statement at a time. If more than 1 SQL statement is needed to be executed it will need to be wrapped inside a **BLOCK** or **PROCEDURE**.
- Tasks can be conditionally executed with the use of **WHEN**. If the conditional expression evaluates to **TRUE** the SQL statement will be executed else it will be ignored. **WHEN** is frequently used with Streams to update the target table when there is data in the stream.
- We can create a Tree of Tasks also called DAG with a root task and several child Tasks.
- We can use the **AFTER** keyword to create parent child relationship between Tasks.
- A Task can have a maximum of 100 children and maximum of 100 parents.
- A DAG can have a maximum of 1000 Tasks including the root Task.

Summary

- Task dependencies can be located with `INFORMATION_SCHEMA.TASK_DEPENDENTS`.
- When we use `RECURSIVE=TRUE` with `TASK_DEPENDENTS` we get child ,grandchild and information on all dependent Tasks in the tree.
- Tasks can be monitored by using the `INFORMATION_SCHEMA.TASK_HISTORY`.
- We have data for 7 days in `INFORMATION_SCHEMA.TASK_HISTORY` and older data can be obtained from `ACCOUNT_USAGE.TASK_HISTORY` which has data for 365 days.
- There is a 45 min lag between Task execution and data showing up in `ACCOUNT_USAGE.TASK_HISTORY`.
- Use `ERROR_ONLY=TRUE` to get Tasks in error state from `TASK_HISTORY`.
- Finalize Task is a Task that is associated with the root Task and runs when all the Tasks in a DAG have completed execution.
- Tasks are commonly used with `MERGE` and `COPY` command to process data in Snowflake.
- We can see execution information on Tasks in Snowsight and visually see dependencies between them.
- Task can be dropped by using `DROP TASK <task-name>`.
- Tasks are automatically suspended when a Task fails 10 times this value can be changed by updating the `SUSPEND_TASK_AFTER_NUM_FAILURES` parameter.
- Tasks do not allow overlapping executions by default, this default behavior can be changed by updating the `ALLOW_OVERLAPPING_EXECUTION` parameter.
- Using `SHOW TASKS` provides us with a list of Tasks.

Thank you



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